VICTORY 2027

Chemical coordination in plants

LECTURE-01 | BIOLOGY

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TOPICS TO BE COVERED

- Plant Hormones


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

Stimuli and Response

- **Stimuli** is a change in either external or internal environment of an organism.
- **Response** is the resulting actions or movements caused by stimuli.

- Plants respond to stimuli by producing chemical compounds called "hormones" that work as messengers.
- The term hormone was first used by William Bayliss and Ernest Starling in 1902.
- Plants are sensitive to light and respond to the direction of light. Also sensitive to water, temperature, gravity, etc.



The plant hormones responsible for chemical coordination in plants are Phytohormones.

Phytohormones

Growth Promoters

- Auxin
  - Natural e.g IAA, IBA
  - Synthetic auxin
    - NAA Naphthalene acetic acid (Read only)
- Gibberlins
- Cytokinin

Growth Inhibitors

- Ethylene
- Abcisic acid [ABA]

Plant Hormones

GO WENT GONE

1. Auxins :



- The term Auxin was first coined by F.W. Went in 1928 from the Greek word auxin which means 'to grow'.

auxein

- This is more effective at low concentrations.
- It promotes cell elongation, which increases cell wall.
- Plant cell wall get loosened like rubber when put under tension, rate of cell elongation is equal to auxin concentration.



Plant Hormones

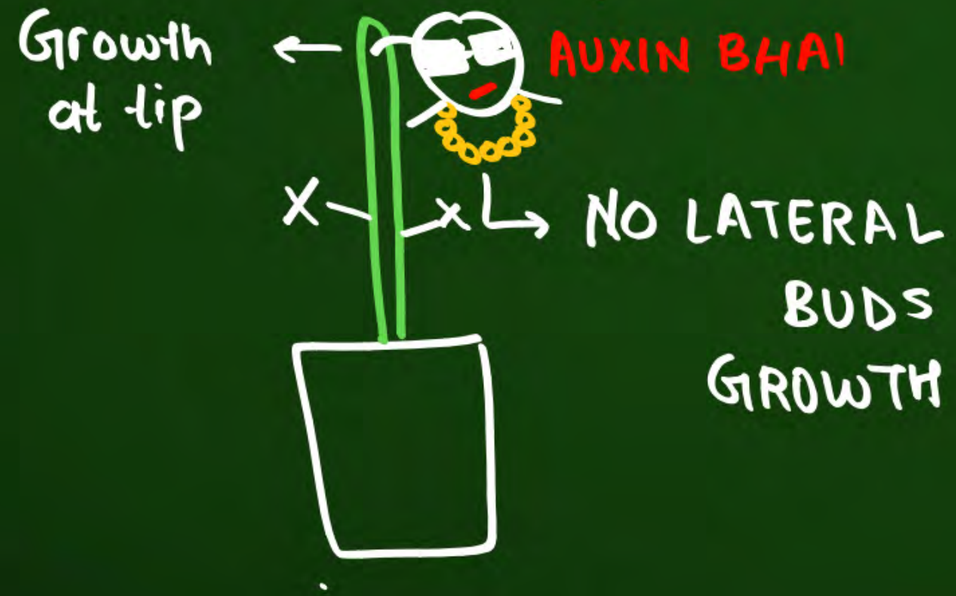
- First growth hormone discovered in plants.
- Are powerful growth-stimulants and also effective at extremely low concentration.
- IAA (Indole-3-acetic acid) main natural auxin in plants.
- Present in shoot apex, root apex, lateral meristems, etc.

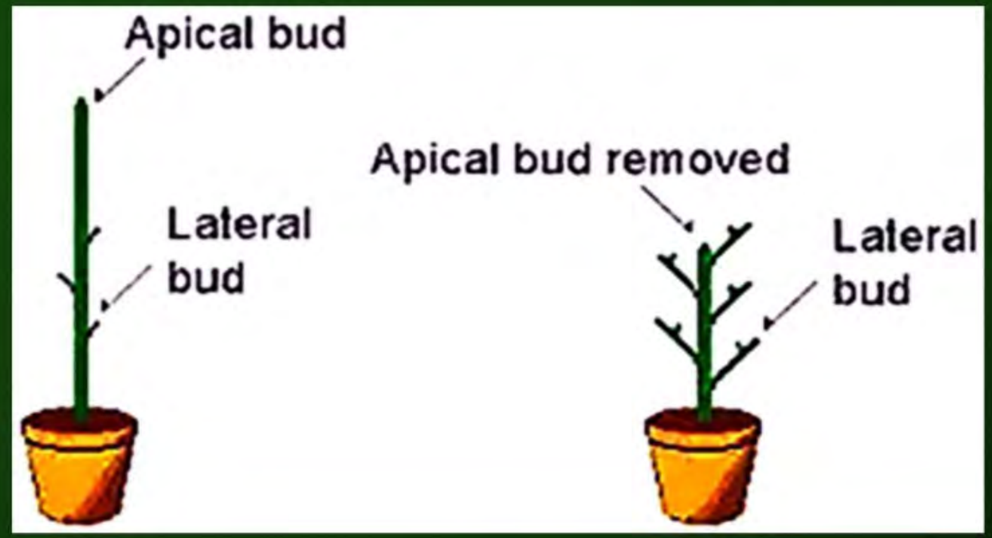


Plant Hormones

❑ Apical Dominance

- The phenomenon of the suppression of the growth of lateral buds by apical buds is called Apical Dominance.



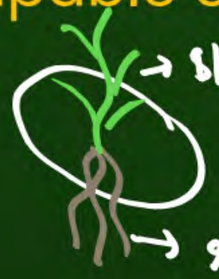


Plant Hormones

2. Gibberellins :

Gibberellic Acid GA,

- Different forms of gibberellins from different plants – GA_1, GA_2, GA_3 and so on.
- Gibberellic acid (GA_3) is most studied.
- Promote growth of internodes by cell elongation.
- Present in meristematic region like stem-apex, root-apex, buds, seeds, etc.
- Breaks seed dormancy (inactive state of seed) and initiate germination.
- Promote fruit growth and are capable of inducing parthenocarpy.
- Delay senescence (ageing).



seed germination

V. imp Term

Parthenocarpy → When there is no fertilisation or no fusion of male & female gamete.

New plant develops without fertilization.

E.g → BANANA

Gibberlin ___

A) Growth Promotors

B) Growth Inhibitor

Gibberlin

Site of Synthesis

Meristem of apical buds, root tips and developing seeds.

Bolting → Internode elongation by cell elongation before flowering. (e.g beet, Cabbage)



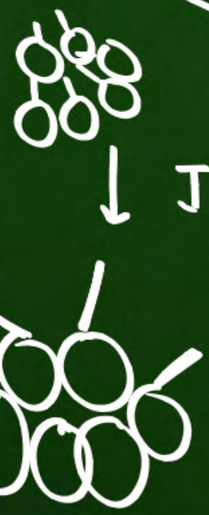
Commercial Use → Horticulture. food industries, Malting Brewing Industry

Wheat —GA→ Beer

Vineyard

① Size of Fruit Increases (Apple)





Internodal Elongation





Plant Hormones

- Causes elongation of stem

Gibberellin



Choose odd one out-

A) CytoKinin

B) Adrenaline

C) Ethylene

D) Auxin

2020

Question

Indole-3-acetic acid is the most common naturally occurring plant hormone of _____ class.

A Gibberellin

B Auxin

C Ethylene

D Cytokinin

Thank You